Steel Plant Energy Forecasting — Methodology Summary

Summary

We forecast the plant's energy use for the next 15-minute interval with **RMSE 8.57 kWh and MAE 4.06 kWh** on unseen future weeks, 33% better than the naive "next = now" forecast (RMSE 12.79). The model averages four gradient-boosted tree models (two LightGBM, XGBoost, CatBoost) using 43 features built only from data up to the current interval.

Key findings from EDA

- **Two operating states:** idle (about 3–5 kWh) or running (40–150 kWh). The hard part is predicting the switch between them.
- **Strong routine:** machines start around 08:00, pause at 12:00 for lunch, stop around 21:00. Weekdays average about 34 kWh, Saturday 16, Sunday 7.
- **Clean data with one quirk:** no missing values and one reading every 15 minutes, but each midnight reading carries the previous day's date; we correct it before building time features.

Features and engineering decisions

43 features in five groups, built by one `make_clues()` function shared by training and inference:

Group	Examples	Signal
Raw readings & calendar (11)	usage, reactive power, CO2, day, load type, month	What the plant is doing now
Time of next interval (5)	15-min slot, hour, sin/cos of clock	Shift start or lunch break coming
Recent usage (18)	lags 1–12, changes, 1h/2h/4h/24h mean and spread	Rising, falling or jumpy
Same moment earlier (4)	next slot yesterday/last week, 4-week max	What usually happens now
Electrical (5)	apparent power (kVA), reactive and power-factor changes	How hard machines work

- **Most valuable groups** (RMSE when removed): time of next interval 8.91, electrical 8.82, recent usage 8.75.
- **Why larger models only work with many features:** with 9 features, larger trees overfit (RMSE 8.96 vs 8.90). With 43 features, slower learning (0.03), more trees (400) and bigger trees (63 leaves) pay off.

How we got here: experiment log

Every change was tested on the same 5 future periods and kept only if average RMSE improved.

Step	Change	RMSE	Kept?
1	Baseline: next = now	12.79	reference
2	LightGBM, 5 features (usage, lag 1, 1-hour average, hour, weekend)	9.64	✓
3	+ load type, + 15-min slot of the next interval	8.99	✓
4	+ change since last reading	9.02	✗
5	+ 24-hour average, + 4-hour maximum (9 features)	8.90	✓
6	Team of 3 models (LightGBM, XGBoost, CatBoost) on 9 features	8.86	✓
7	Single extra features on the 9 (CO2, lag 2, yesterday's jump, …)	8.95–9.13	✗
8	Larger models on the 9 features	8.96	✗
9	Other targets: predict the change, or log of usage	8.90 / 8.95	✗
10	**43 features + larger models + team of 4**	**8.57**	✓ final

Two lessons: "change since last reading" looked helpful on a single test period (8.38 → 8.30) but hurt on 5 periods, so one test period can mislead; and small additions to a small feature set failed while the full feature set with larger models succeeded.

Validation approach and scores

Expanding-window **TimeSeriesSplit, 5 folds**: each trains on the past and tests on the next 4 weeks (2,688 intervals). Random K-fold was rejected because it trains on future data.

Model	RMSE (mean ± std)
Team of 4 (final)	**8.57 ± 0.40**
CatBoost / XGBoost / LightGBM alone	8.64 / 8.65 / 8.71
Baseline: next = now	12.79 ± 0.72

A dress rehearsal of the official procedure (retrain without the last 4 weeks, then call the inference function with only the 11 feature columns) gives **RMSE 8.55, MAE 3.80** (baseline 12.37 / 5.43). Storing the last 7 days of training data as history matters here: without it RMSE is 8.94. R^2 is 0.92 and the running/idle state is right 98.7% of the time.

Main limitation: jumps of more than 20 kWh are 10% of intervals but 64% of the squared error. A machine start is a human decision the telemetry cannot reveal in advance.

Business takeaways and interpretability

- **Alarm with a margin to avoid penalties.** With a penalty threshold at 100.8 kWh (95th percentile), alarming when the forecast exceeds 80 kWh catches **81% of breaches** one interval ahead (precision 35%), versus 73% (precision 32%) when alarming on the current reading.
- **Explainable drivers:** apparent power (54% of LightGBM gain) and current usage (29%) dominate, followed by CO2, the 4-week maximum at this slot and yesterday's jump. In plain words: "current electrical load, adjusted by the plant's routine".
- **Recommendation:** watch the 08:00 start-up and 13:00 restart, where errors peak. Logging planned production schedules would be the single biggest improvement.